

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route
ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

This output indicates that R3 does not have a direct OSPF neighbor relationship with R2. Consequently, R3 must send all traffic destined for the 192.168.2.0/24 network through its other neighbor, R1.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf

1

R2(config-router)# no passive-interface

s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 This is calculated by adding the OSPF cost of the outgoing interface on R3 (Serial0/0/1, which is a T1 link at 1.544Mb/s with a cost of 64) to the cost of the interface on R2 that connects to the destination network (GigabitEthernet0/0, which has a cost of 1).

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth of 100 Mbps is insufficient for modern high-speed links (like GigabitEthernet), as it would assign them the same cost as FastEthernet. Increasing the reference bandwidth allows for more granular and accurate cost calculations, ensuring that faster interfaces are correctly preferred by the routing protocol.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

- 192.168.3.0/24: The path from R1 to this network goes through R1's S0/0/1 interface and then R3's G0/0 interface. With the changed bandwidth settings, the cost on R1's serial interface was 781, and the cost on R3's GigabitEthernet interface was 1, resulting in a total accumulated cost of $781 + 1 = 782$.

- 192.168.23.0/30: This network is directly connected to R3's S0/0/1 interface. The path from R1 goes out its S0/0/1 interface (cost 781) and into R3's S0/0/1 interface (cost 64), for a total of $781 + 64 = 845$.

Lab - Configuring Basic Single-Area OSPFv2

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. This is because the bandwidth 128 command sets the serial interfaces to 128 Kbps. The OSPF cost calculation (Reference Bandwidth / Interface Bandwidth) with a reference bandwidth of 100,000 Kbps results in a cost of 781 for each 128 Kbps serial link. The path from R1 to the 192.168.23.0/24 network now traverses two such serial links (R1->R2 and R2->R3), so the total cost is $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

After manipulating the interface bandwidths or costs, the path via R2 now has a lower total OSPF cost compared to the direct path from R1 to R3. OSPF always installs the route with the lowest metric (shortest path) to a destination in the routing table, which in this case is the path through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Controlling the Router ID (RID) manually ensures stability and predictability within the OSPF domain. A manually assigned RID prevents the router from having to re-elect a Designated Router (DR) and Backup Designated Router (BDR) or potentially change its OSPF identity if the interface used for the automatically selected RID goes down or is reconfigured, which could disrupt routing.

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR election is only relevant on multi-access broadcast networks (like Ethernet) where multiple routers are connected to the same segment. In this lab, all connections between routers are point-to-point serial links.

3. Why would you want to set an OSPF interface to passive?

____ Configuring an interface as passive in OSPF serves two main purposes: it prevents the router from sending OSPF hello messages out of that interface, which ~~conserves bandwidth and processing resources on links where no other OSPF routers exist~~. Simultaneously, it allows the network connected to that passive interface to still be advertised as an OSPF route to other routers.

4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

____ The use of "areas" is the key feature that gives OSPF its hierarchical design. By dividing a large network into multiple areas, routing updates and topology information can be contained within each area. This significantly reduces the size of the link-state database on routers, lowers processing overhead, and minimizes routing traffic, enabling OSPF to scale to very large network sizes.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

____ network 10.0.0.0 0.255.255.255 area 0

4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

____ The ~~OSPF~~ route will be installed in the routing table and used for forwarding. Although its metric (cost) is much higher than RIP's hop count, the routing decision is primarily based on Administrative Distance (AD). OSPF has a default AD of 110, which is lower than RIP's AD of 120 and EIGRP's internal AD of 90.

6